

Notes: Henry (JF, 2000)
**Stock Market Liberalization, Economic Reform, and Emerging Market Equity
Prices**

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Countries and stock-market data	4
3.2	Liberalization dates	4
3.3	Announcement dates	5
3.4	Economic reform controls	5
3.5	Macroeconomic fundamentals	5
4	Methodology and empirical specifications	5
4.1	Baseline event-study specification	5
4.2	World stock-return controls	6
4.3	Concurrent reform controls	6
4.4	Macroeconomic fundamentals	6
4.5	Alternative event windows	6
4.6	Subsequent liberalizations	7
5	Identification and interpretation	7
5.1	What the paper can identify	7
5.2	Why the window starts before implementation	7
5.3	Core identifying assumption	7
5.4	Policy endogeneity	7
5.5	Interpreting the main coefficient	8
5.6	Cost-of-capital interpretation	8
5.7	What remains ambiguous	8
6	Tables and figures	8
6.1	Tables I and II: Liberalization dates and date comparisons	8

6.2	Figure 1: Returns and dividend yields around liberalization	9
6.3	Table III: Announcement dates	10
6.4	Table IV: Liberalizations and contemporaneous economic reforms	11
6.5	Tables V and VI: Main return results and alternative window lengths	11
6.6	Appendix Table AI: Subsequent liberalizations and reform context	12
6.7	Appendix Tables AIII and AIV: Alternative dates and later liberalizations	14
7	Short summary	15
8	Conclusion	15
8.1	Core conclusion	15
8.2	Economic intuition	16
8.3	Incremental contribution revisited	16
8.4	Implications	16
8.5	Limitations	16
8.6	Takeaway	16

1 Big picture

- **Research question.** Does a country's initial stock market liberalization raise aggregate equity prices in emerging markets? Equivalently, is the stock market's reaction consistent with a fall in the country's cost of equity capital when foreigners are allowed to buy domestic shares?
- **Definition.** A stock market liberalization is a government decision to allow foreign investors to purchase shares in the domestic stock market.
- **Main theory.** Standard international asset pricing models predict that liberalization can reduce the equity premium by moving the market from domestic risk pricing toward world risk pricing. In a segmented market, the relevant risk is closer to the variance of domestic cash flows; in an integrated market, the relevant risk is closer to the covariance with the world portfolio.
- **Core price implication.** Holding expected future cash flows fixed, a lower equity premium implies a lower discount rate and therefore a higher aggregate equity price index.
- **Main finding.** In the eight-month window from $T^* - 7$ through T^* , where T^* is the implementation month of the first liberalization, emerging market indices earn large abnormal returns. The benchmark estimate is about 4.7 percent per month. After controlling for world returns, economic reforms, and macroeconomic fundamentals, the estimate is still about 3.3 percent per month, or roughly 26 percent cumulatively.
- **Robustness.** Shorter windows give positive estimates as well. The implementation-month-only estimate is about 6.5 percent, which weakens the concern that the result is only an artifact of a long pre-event run-up.
- **Important caveat.** Liberalizations often occur at the same time as trade reforms, privatizations, stabilization plans, and exchange-control reforms. The paper is careful about these confounds, but it still emphasizes that causal interpretation requires caution.

Economic takeaway. Stock market liberalization appears to be associated with a large revaluation of emerging market equity prices. The evidence is consistent with a lower cost of equity capital, but the event is embedded in broader reform packages, so the estimates should be read as carefully controlled event-study evidence rather than a perfectly isolated natural experiment.

2 Incremental contribution of the paper

- **From integration to valuation.** Earlier work studied whether emerging markets become more integrated with global markets after liberalization. Henry focuses directly on the valuation effect: do prices jump when markets open?
- **Careful event dating.** The paper constructs a systematic list of first liberalization dates using official policy decrees, first country-fund introductions, and large jumps in the IFC investability index.
- **Announcement-date problem.** The paper shows that news-database announcement dates are often unreliable or occur after implementation. This justifies using a window before implementation rather than only the first newspaper announcement.

- **Explicit reform controls.** A major contribution is the construction of a country-by-country dataset of macroeconomic reforms around liberalization. The paper controls separately for stabilization, trade liberalization, privatization, and easing of exchange controls.
- **Stock-market response to macro reforms.** The paper also estimates how equity prices respond to nonfinancial reforms. Trade liberalization and privatization are themselves associated with positive stock-market revaluations.
- **Window robustness.** The paper reports eight-month, five-month, two-month, and one-month event windows. The one-month implementation result is still positive and significant.
- **First versus later liberalizations.** Later liberalizations produce much smaller and statistically insignificant revaluations, suggesting that the initial opening contains most of the market-moving news.

3 Data and measurement

3.1 Countries and stock-market data

The sample contains 12 emerging markets: Argentina, Brazil, Chile, Colombia, Mexico, and Venezuela in Latin America, and India, Malaysia, Korea, the Philippines, Taiwan, and Thailand in Asia.

- **Source.** Emerging-market returns come from the IFC Emerging Markets Data Base.
- **Return measure.** The country return is the IFC Total Return Index, denominated in U.S. dollars and deflated by the U.S. CPI.
- **Frequency.** The analysis uses monthly data and logarithmic returns.
- **World-market controls.** The main controls are the IFC global emerging-market index, the S&P 500, and the Morgan Stanley Europe, Asia, and Far East index.
- **Dividend yields.** The paper also studies changes in the log dividend yield, $\Delta \ln(D/P)$, because a price revaluation should show up as a fall in dividend yields if dividends do not rise mechanically at the same time.

3.2 Liberalization dates

Henry defines the first stock market liberalization as the first month in which one of the following occurs:

- a verifiable policy decree allows foreign portfolio investment;
- the first country fund is introduced, which indirectly allows foreign ownership of domestic equities;
- the IFC investability index increases by at least 10 percent.

Interpretation. This dating procedure is designed to identify the first economically meaningful opening, not necessarily the date of a complete or final opening. Many countries liberalize gradually, but the first opening is likely to carry the largest news about future access.

3.3 Announcement dates

The paper searches Lexis/Nexis for announcement dates using country-fund names and phrases such as stock market liberalization, stock market opening, capital market liberalization, foreign investment, and foreign portfolio investment.

- For many countries, the earliest news found appears in or after the implementation month.
- Several apparent announcements are not true stock-market announcements; for example, Colombia’s February 1991 item is about trade liberalization rather than stock market liberalization.
- Because the announcement dates are not credible first-information dates, the paper uses an event window that starts before implementation.

3.4 Economic reform controls

The paper codes four types of reforms that could also affect expected cash flows and equity prices:

- **Stabilize:** macroeconomic stabilization programs, including IMF agreements and anti-inflation plans;
- **Trade:** trade liberalization and tariff/import restrictions reforms;
- **Privatize:** privatization programs and sales of state enterprises;
- **Exchange:** easing of exchange controls and capital-flow restrictions outside the stock-market opening itself.

Each reform dummy uses the same basic window logic as the stock-market liberalization dummy: it turns on in the months leading up to the reform implementation.

3.5 Macroeconomic fundamentals

To control for country fundamentals and possible policy timing, the paper also includes lags, contemporaneous values, and leads of growth rates in fundamentals such as domestic industrial production, domestic inflation, the real exchange rate, the U.S. Treasury bill rate, and a political-stability index.

4 Methodology and empirical specifications

4.1 Baseline event-study specification

The baseline panel regression is:

$$R_{it} = \alpha_i + \gamma Liberalize_{it} + \varepsilon_{it}, \quad (1)$$

where R_{it} is the real dollar return for country i in month t , α_i are country fixed effects, and $Liberalize_{it}$ equals one during the eight-month first-liberalization window $T^* - 7$ through T^* .

- γ is the average monthly abnormal return during the liberalization window.
- The cumulative effect is approximately $8 \times \gamma$ for the eight-month window because returns are log returns.
- Country fixed effects remove average return differences across countries.

4.2 World stock-return controls

The first augmented specification adds global stock-return controls:

$$R_{it} = \alpha_i + \beta_1 R_t^{LDC} + \beta_2 R_t^{US} + \beta_3 R_t^{EAFE} + \gamma Liberalize_{it} + \varepsilon_{it}. \quad (2)$$

- R_t^{LDC} captures broad emerging-market movements.
- R_t^{US} captures U.S. market movements.
- R_t^{EAFE} captures developed-market movements outside the United States.
- If the liberalization result were just a global equity boom, the coefficient on *Liberalize* should fall sharply after adding these controls.

4.3 Concurrent reform controls

The next specification adds dummies for other economic reforms:

$$R_{it} = \alpha_i + \beta_1 R_t^{LDC} + \beta_2 R_t^{US} + \beta_3 R_t^{EAFE} + \gamma_1 Liberalize_{it} \quad (3)$$

$$+ \gamma_2 Stabilize_{it} + \gamma_3 Trade_{it} + \gamma_4 Privatize_{it} + \gamma_5 Exchange_{it} + \varepsilon_{it}. \quad (4)$$

Interpretation. These controls are essential because a country often liberalizes its stock market as part of a broad reform package. Without them, a stock price response to trade reform or privatization could be misattributed to stock market liberalization.

4.4 Macroeconomic fundamentals

The most complete specification adds distributed lags and leads of macroeconomic fundamentals:

$$R_{it} = \alpha_i + \beta_1 R_t^{LDC} + \beta_2 R_t^{US} + \beta_3 R_t^{EAFE} + \gamma_1 Liberalize_{it} \quad (5)$$

$$+ \gamma_2 Stabilize_{it} + \gamma_3 Trade_{it} + \gamma_4 Privatize_{it} + \gamma_5 Exchange_{it} + \delta(L)\Delta \ln(F_{it}) + \varepsilon_{it}. \quad (6)$$

The point of this specification is to ask whether the liberalization dummy is simply proxying for good macroeconomic news.

4.5 Alternative event windows

The benchmark window is $T^* - 7$ to T^* , but the paper also estimates:

- a five-month window, $T^* - 4$ to T^* ;

- a two-month window, $T^* - 1$ to T^* ;
- an implementation-month-only window, T^* .

Reason. Shorter windows address the concern that policymakers may liberalize after a long stock-market run-up. If the estimate remains positive in the implementation month, the result is less likely to be only a market-timing artifact.

4.6 Subsequent liberalizations

The paper separates the initial liberalization from later liberalizations by defining another dummy, *Liberalize2*, for implementation months of stock market liberalizations that occur after the first.

Interpretation. If the first opening reveals the future path of liberalization, later openings should contain less new information. That is exactly what the paper finds.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies abnormal stock-market performance around first stock-market liberalizations after controlling for world market returns, other reforms, and macroeconomic fundamentals. The design is a panel event study with many controls, not a randomized policy experiment.

5.2 Why the window starts before implementation

The paper does not rely on newspaper announcement dates because those dates are often late, incomplete, or contaminated by non-stock-market reforms. The pre-implementation window captures gradual information leakage and expectations formation.

5.3 Core identifying assumption

The theory says prices should rise because the discount rate falls, holding expected future cash flows constant. Empirically, the paper tries to hold expected cash flows constant by controlling for:

- other macroeconomic reforms that may raise future profitability;
- comovement with global stock markets;
- macroeconomic fundamentals and their near-term dynamics.

5.4 Policy endogeneity

A major threat is policy timing. Governments may open markets when stock prices are already rising or when economic conditions are expected to improve. The paper addresses this by adding controls and shorter windows, but it does not claim that all endogeneity is eliminated.

5.5 Interpreting the main coefficient

In the full eight-month specification, the coefficient on *Liberalize* is about 0.033. Because the window lasts eight months, the cumulative revaluation is approximately:

$$8 \times 0.033 \approx 0.264,$$

which means about 26 percent in log-return terms.

In the implementation-month-only specification, the coefficient is about 0.065, or about 6.5 percent in that month.

5.6 Cost-of-capital interpretation

If expected future cash flows are fixed, a price increase maps into a fall in the discount rate. Henry gives a simple calculation: if the preliberalization discount rate is 20 percent, then a 26 percent reduction in the cost of equity lowers the discount rate to roughly 15 percent. The more conservative one-month estimate lowers it to roughly 19 percent.

5.7 What remains ambiguous

The paper emphasizes that stock market liberalization may also be interpreted by investors as a signal of future reforms. For example, trade reforms may signal future stock-market openings, and stock-market openings may signal future macro reforms. This sequencing problem makes exact causal attribution difficult.

6 Tables and figures

6.1 Tables I and II: Liberalization dates and date comparisons

Read:

- Table I lists the first stock market liberalization date for each of the 12 countries and explains whether the opening came through a policy decree, a country fund, or a jump in the investability index.
- Several first openings occur through country funds rather than formal complete legal openings.
- Table II compares Henry's dates with Bekaert and Harvey, Kim and Singal, and Buckberg.
- The paper uses the earliest verifiable stock-market liberalization date from Henry's procedure as the benchmark.

Economic message: Event-study estimates are only as credible as the event dates. These tables show that the paper is not simply accepting one existing date list; it constructs and defends the dates used for the main analysis.

Table I
First Stock Market Liberalization

The stock market liberalization dates are based on information obtained from the following sources: Levine and Zervos (1994); *The Wilson Directory of Emerging Market Funds*; IFC Investable Indices; Park and Van Agtmael (1993); Price (1994); *The Economist Intelligence Unit*, various issues; *The Economist Guide to World Stock Markets* (1988); and the IMF's *Exchange Arrangements and Restrictions*, various issues.

Country	Date of First Stock Market Liberalization	Details about the Liberalization
Argentina	November 1989	Policy Decree: The liberalization began with the New Foreign Investment Regime in November 1989. Legal limits on the type and nature of foreign investments are reduced (Park and Van Agtmael (1993), p. 326).
Brazil	March 1988	Country Fund Introduction: "The Brazil Fund Incorporated" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 17).
Chile	May 1987	Country Fund Introduction: "The Toronto Trust Mutual Fund" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 17).
Colombia	December 1991	Policy Decree: Resolution 52 allowed foreign investors to purchase up to 100 percent of locally listed companies (Price (1994)).
India	June 1986	Country Fund Introduction: "The India Fund" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 12).
Korea	June 1987	Country Fund Introduction: "The Korea Europe Fund Limited" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 13).
Malaysia	May 1987	Country Fund Introduction: "The Wardley GS Malaysia Fund" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 14).
Mexico	May 1989	Policy Decree: Restrictions on foreign portfolio inflows were substantially liberalized (Levine and Zervos (1994)).
The Philippines	May 1986	Country Fund Introduction: "The Thornton Philippines Redevelopment Fund Limited" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 15).
Taiwan	May 1986	Country Fund Introduction: "The Taipei Fund" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 15).
Thailand	January 1988	Country Fund Introduction: "The Siam Fund Limited" (<i>The Wilson Directory of Emerging Market Funds</i> , p. 16).
Venezuela	January 1990	Policy Decree: Decree 727 completely opened the market to foreign investors except for bank stocks ((Levine and Zervos (1994)).

Table I: First stock market liberalization dates.

Table II
Comparison of Official Liberalization Dates across Authors

The dates in column (2) are constructed using the dating procedure described in the paper. The dates in columns (3) through (5) are taken from Bekaert and Harvey (2000), Kim and Singal (2000), and Buckberg (1995), respectively. Column 6 shows the earliest date given for a country in the preceding four columns.

(1) Country	(2) Dating Procedure	(3) Bekaert & Harvey	(4) Kim & Singal	(5) Buckberg	(6) Earliest
Argentina	11-89	11-89	11-89	10-91	11-89
Brazil	3-88	5-91	5-91	5-91	3-88
Chile	5-87	1-92	9-87	10-89	5-87
Colombia	12-91	2-91	2-91	10-91	2-91
India	6-86	11-92	11-92	NA	6-86
Korea	6-87	1-92	1-92	NA	6-87
Malaysia	5-87	12-88	12-88	NA	5-87
Mexico	5-89	5-89	11-89	5-89	5-89
The Philippines	5-86	6-91	7-86	10-89	5-86
Taiwan	5-86	1-91	1-91	NA	5-86
Thailand	1-88	9-87	8-88	NA	9-87
Venezuela	1-90	1-90	1-90	12-88	12-88

Table II: Comparison of official liberalization dates across authors.

6.2 Figure 1: Returns and dividend yields around liberalization

Read:

- The upward-sloping line plots cumulative abnormal returns around the first stock market liberalization.
- The downward-sloping line plots cumulative abnormal changes in log dividend yields.
- From roughly $T^* - 12$ to T^* , returns rise substantially while dividend yields fall.
- The mirror-image pattern is consistent with price revaluation rather than a mechanical dividend effect.

Economic message: This is the visual motivation for the paper. Equity prices rise and dividend yields fall before liberalization, which is exactly the pattern expected if the cost of equity capital declines.

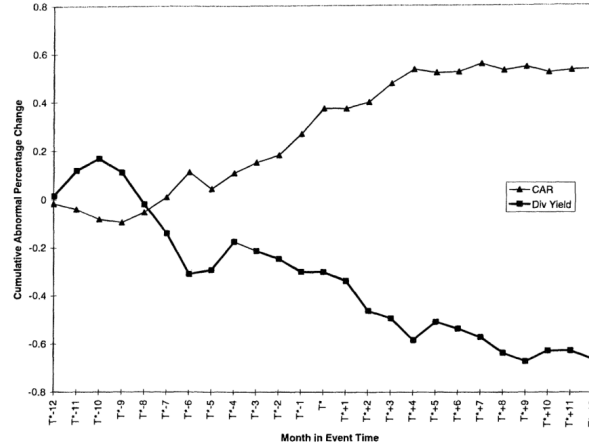


Figure 1. The behavior of stock returns and dividend yields around the first stock market liberalization. The variable on the y-axis is the continuously compounded abnormal percentage change. T^* is the month in which the stock market liberalization was implemented. The upward trending series (triangles) is a plot of the cumulative residuals from a panel regression of the real dollar return from all 12 countries on a constant and 11 country-specific dummies. The downward trending series (squares) is a plot of the cumulative residuals from a panel regression of the change in the natural log of the dividend yield on a constant and 11 country-specific dummies.

Figure 1: Behavior of stock returns and dividend yields around the first stock market liberalization.

6.3 Table III: Announcement dates

Read:

- Table III reports the Lexis/Nexis search results for announcement dates.
- In many countries, the first news item appears in or after the implementation month.
- Some news items are about related reforms rather than actual stock-market liberalization.
- The evidence supports the paper's decision not to rely mechanically on newspaper announcements.

Economic message: The relevant market-information date is difficult to observe. The event window is meant to capture leaked information and gradual learning before the formal implementation date.

(1) Country and Implementation Date	(2) Announcement Date(s)	(3) Source	(4) Headline
Argentina (November 1989)	December 11, 1989	<i>The Financial Times</i>	Argentina fund aims at privatised companies.
Brazil (March 1988)	March 23, 1988 March 31, 1988 April 4, 1988	<i>The Toronto Financial Post</i> PR Newswire Institutional Investor, Inc.	Some like it hot: Shares in the fund will be offered to the public shortly by first Boston Corporation and Merrill Lynch Capital Markets Brazil Fund Common Stock Offered Brazil Fund is Hot
Chile (May 1987)	February 7, 1996	<i>The Reuter European Business Report</i>	Micropal names best 1995 emerging market funds. The Toronto Trust Chile Fund, launched in 1987, is Micropal's best performing emerging market fund over the past seven years
Colombia (December 1991)	February, 1991	National Trade Data Bank Market reports	Colombia-Economic Policy and Trade Practices. The administration of President Gaviria has embarked on "la apertura" (the opening), a bold plan to lower tariffs and other barriers to foreign trade
India (June 1986)	May 12, 1986 June 17, 1986	<i>The Financial Times</i> <i>The Financial Times</i>	Maverick Brings in the Savings. The government approved the Unit trust of India's (UTI) collaboration with Merrill Lynch to launch the India Fund More Details Given for India Fund. The Indian government last week approved the proposal which for the first time will allow foreigners to invest in the Indian stock markets

Table III, part 1: Announcement dates for first stock market liberalizations.

Korea (June 1987)	March 21, 1987	<i>The Economist</i>	South Korean Securities; Authorised Entry Only
Malaysia (May 1987)	April 8, 1987 May 11, 1987	Jiji Press Limited U.P.I.	Arab-Malaysian Merchant Bank—IFC Move to Tap U.S. Market. Malaysian Fund Offering Increased
Mexico (May 1989)	May 15, 1989 July 8, 1989	Reuters <i>The New York Times</i>	Mexico Announces New Foreign Investment Rules Mexico Eases Foreign Curb. The government has opened Mexico's stock exchange to foreign investment
Philippines (May 1986)	September 22, 1986	<i>Business Week</i>	For Aquino, U.S. Business Will Be a Tough Sell. Tai: Hong Kong-based Thornton Management (Asia) Ltd. recently launched the Philippines Redevelopment Fund which invests in Philippine stocks
Taiwan (May 1986)	July 3, 1985 June 28, 1986	Central News Agency <i>The Economist</i>	Local Securities Investment Company Formed in Taipei Details: A 25 million dollar investment fund to be called the Taipei Fund will be raised soon Asian Funds Details: The Taipei Fund was formed on May 22 nd
Thailand (January 1988)	April 27, 1988	<i>The Financial Times</i>	Another Thai Fund to Join the Market Details: the fund was established in January
Venezuela (January 1990)	December, 1989	<i>South Magazine</i>	Scramble at the Fringe: Third World Stock Markets Details: Liberalisation is proceeding in Argentina and Venezuela

Table III, part 2: Continued announcement dates.

6.4 Table IV: Liberalizations and contemporaneous economic reforms

Read:

- Table IV lists major economic events around each first stock market liberalization.
- The table documents many overlapping reforms: tariff reductions, privatization, IMF agreements, exchange-rate stabilization, and exchange-control changes.
- Argentina, Colombia, the Philippines, Malaysia, and Venezuela illustrate how hard it is to isolate stock-market opening from broader reform packages.

Economic message: This table is the central reason the paper needs reform controls. Stock market liberalization often arrives as part of a larger reform sequence, so the raw event-study return is not automatically a pure cost-of-capital effect.

Country, Date of Liberalization	Type of Liberalization	Event Time					
		$T^* - 12$	$T^* - 9$	$T^* - 6$	$T^* - 3$	T^*	$T^* + 3$
Argentina November 1989	Limits on foreign capital reduced	Airline privatization; dual exchange rate system fails	Structural adjustment funds frozen; economic team resigns	Privatization stabilization plan	IMF agreement	Exchange rate devalued by 35 percent	IMF agreement frozen
Brazil March 1988	Country Fund	Finance minister resigns	Second Cruzado Plan	New proposals submitted to creditors	None	Capital goods duties reduced	Tariffs reduced
Chile May 1987	Country Fund	None	Attempt on Pinochet's life	None	Largest banks privatized; new debt repayment terms	None	Two floods and an earthquake
Colombia December 1991	Investability index jumps 46 percent	Restrictions on profit remittance eased	Tariffs reduced; external debt refinanced	Tariffs cut; credit controls relaxed	Exchange controls eased	Privatization of telecom industry begins	None
India June 1986	Country Fund	None	None	None	None	None	Attempt on Prime Minister's life
Korea June 1987	Country Fund	None	None	None	None	None	False rumors of Kim Il Sung's death
Malaysia February 1987	Country Fund	None	None	None	None	National Economic Plan (NEP) frozen	NEP to be extended past 1990
Mexico May 1989	Investability index jumps 410 percent	Salinas elected; U.S. govt. gives \$3.5B to boost reforms	Pact extended	Privatization of two state mines	Brady Plan approved by U.S. Congress; IMF agreement	None	Brady agreement with creditors
The Philippines May 1986	Country Fund	Debt rescheduling signed	IMF targets missed	\$ 2.9 billion of public debt rescheduled	Marcos overthrown	Import restrictions lifted	Talks open with IMF
Taiwan May 1986	Country Fund	None	None	Investment in foreign securities allowed	None	Import bans lifted	Exchange controls eased
Thailand January 1988	Country Fund	General Yongchaiyut calls for reforms	None	ASEAN free trade agreement extended	None	None	None
Venezuela January 1990	Full market access except bank stocks	Trade liberalization; adjustment loan approved	None	None	Easier profit remittance for foreign firms	\$680 million structural adjustment loan	Brady deal; Agricultural tariffs reduced

Table IV, part 1: First liberalizations and contemporaneous economic reforms.

Table IV, part 2: Continued reform events.

6.5 Tables V and VI: Main return results and alternative window lengths

Read Table V:

- Column (1a) gives the simple benchmark return estimate: $Liberalize = 0.047$, implying about 38 percent over eight months.
- Adding world returns lowers the coefficient to 0.041.
- Adding reform controls lowers it to 0.039.
- Adding macro fundamentals lowers it to 0.033, implying about 26 percent over the eight-month window.
- Trade liberalization is also strongly positive in the return regressions and strongly negative in the dividend-yield regressions.

Read Table VI:

- Shorter windows still produce positive liberalization coefficients.
- The five-month estimate is 0.030, the two-month estimate is 0.050, and the implementation-month-only estimate is 0.065.

- The implementation-month-only result is the largest point estimate, which weakens the market-timing critique.

Economic message: The core revaluation result survives many controls and does not disappear when the window is shortened. The main estimate is smaller after controls, which shows that the controls matter, but it remains economically large.

Table V

Stock Market Reactions to First Stock Market Liberalization

The regressions are performed using monthly stock market data from December 1976 to December 1994 for Argentina, Brazil, Chile, India, Korea, Mexico, and Thailand. For the other countries the data are monthly from December 1984 to December 1994. The dividend yield data are also monthly and cover the period from December 1984 to December 1994. *Liberalize* is a dummy variable for the event window of the first stock market liberalization. The event window begins seven months prior to the implementation month and ends in the implementation month. For example, for a stock market liberalization that was implemented in November 1989, the event window begins in April 1989 and ends in November 1989. R^{LDC} , R^{US} , and R^{EAFE} are the dividend-inclusive monthly return on the IFC global index, the S&P 500, and the Morgan Stanley Capital Index for Europe, Asia, and the Far East, respectively. *Stabilize*, *Trade*, *Privatize*, and *Exchange* are dummy variables for the event windows of macroeconomic stabilization, trade opening, privatization, and exchange controls, respectively. Each of the event windows for these economic reform variables begins seven months prior to the implementation of the reform and ends in the implementation month. A constant plus 11 country dummies were also estimated but not reported. Heteroskedasticity-consistent (White) standard errors are in parentheses.

	Panel A: Stock Returns				Panel B: $\Delta \ln(D/P)$			
	(1a)	(2a)	(3a)	(4a)	(1b)	(2b)	(3b)	(4b)
<i>Liberalize</i>	0.047*** (0.010)	0.041*** (0.0124)	0.039*** (0.012)	0.033*** (0.011)	-0.024* (0.015)	-0.019 (0.015)	-0.015 (0.015)	-0.010 (0.017)
R^{LDC}		0.522*** (0.148)	0.517*** (0.015)	0.525*** (0.142)		-0.350*** (0.114)	-0.341*** (0.110)	-0.339*** (0.115)
R^{US}		0.250*** (0.102)	0.278*** (0.109)	0.278*** (0.109)		-0.355* (0.200)	-0.365* (0.205)	-0.446** (0.200)
R^{EAFE}		-0.008 (0.044)	-0.006 (0.044)	-0.018 (0.042)		-0.043** (0.020)	-0.045** (0.022)	-0.027 (0.024)
<i>Stabilize</i>			0.003 (0.010)	0.003 (0.010)			-0.003 (0.010)	0.003 (0.010)
<i>Trade</i>			0.025*** (0.005)	0.021*** (0.048)		-0.039*** (0.015)	-0.037** (0.016)	
<i>Privatize</i>				0.016** (0.007)		-0.029 (0.019)	-0.030 (0.021)	
<i>Exchange</i>				-0.005 (0.015)		-0.002 (0.015)	0.007 (0.049)	
$\bar{R}^2$	0.007	0.076	0.083	0.147	0.000	0.018	0.023	0.027
No. of obs.	2292	2292	2292	2292	1569	1569	1569	1569

Table V: Stock market reactions to first stock market liberalization.

Table VI

Stock Market Reactions to First Stock Market Liberalization, Alternative Event Window Lengths

The regressions are performed using monthly stock market data from December 1976 to December 1994 for Argentina, Brazil, Chile, India, Korea, Mexico, and Thailand. For the other countries the data are monthly from December 1984 to December 1994. The dividend yield data are also monthly and cover the period from December 1984 to December 1994. *Liberalize* is a dummy variable for the event window of the first stock market liberalization. For $T^* - 4$ to T^* , the event window begins four months prior to the implementation month and ends in the implementation month. For example, for a stock market liberalization that was implemented in November 1989, the event window begins in July 1989 and ends in November 1989. For $T^* - 1$ to T^* , the event window begins in the month before the implementation month. For T^* , the event window is the implementation month only. R^{LDC} , R^{US} , and R^{EAFE} are the dividend-inclusive monthly return on the IFC Global Index, the S&P 500, and the Morgan Stanley Capital Index for Europe, Asia, and the Far East, respectively. *Stabilize*, *Trade*, *Privatize*, and *Exchange* are dummy variables for the event window of macroeconomic stabilization, trade opening, privatization, and exchange controls, respectively. Each of the event windows for these economic reform variables begins seven months prior to the implementation of the reform and ends in the implementation month. A constant plus 11 country dummies were also estimated but not reported. Heteroskedasticity-consistent (White) standard errors are in parentheses.

	Panel A: Stock Returns			Panel B: $\Delta \ln(D/P)$		
	$T^* - 4$ to T^*	$T^* - 1$ to T^*	T^*	$T^* - 4$ to T^*	$T^* - 1$ to T^*	T^*
<i>Liberalize</i>	0.030* (0.018)	0.050* (0.028)	0.065* (0.039)	0.017 (0.032)	-0.008 (0.051)	-0.003 (0.076)
R^{LDC}	0.520*** (0.058)	0.522*** (0.058)	0.522*** (0.059)	-0.340*** (0.118)	-0.340*** (0.111)	-0.339*** (0.116)
R^{US}	0.283*** (0.091)	0.280*** (0.091)	0.281*** (0.094)	-0.451** (0.200)	-0.367* (0.204)	-0.448** (0.197)
R^{EAFE}	-0.016 (0.036)	-0.0150 (0.036)	-0.014 (0.033)	-0.028 (0.024)	-0.0276 (0.021)	-0.0281 (0.024)
<i>Stabilize</i>	0.003 (0.010)	0.003 (0.010)	0.003 (0.010)	0.002 (0.008)	0.003 (0.008)	0.003 (0.008)
<i>Trade</i>	0.021** (0.009)	0.020** (0.009)	0.020** (0.009)	-0.037** (0.016)	-0.037** (0.017)	-0.037** (0.017)
<i>Privatize</i>	0.010 (0.009)	0.011 (0.009)	0.011 (0.009)	-0.030 (0.021)	-0.030 (0.021)	-0.030 (0.0210)
<i>Exchange</i>	-0.002 (0.014)	-0.002 (0.014)	-0.003 (0.014)	0.008 (0.045)	0.008 (0.045)	0.008 (0.045)
$\bar{R}^2$	0.146	0.146	0.146	0.027	0.027	0.027
No. of obs.	2292	2292	2292	1569	1569	1569

Table VI: Alternative event window lengths.

6.6 Appendix Table AI: Subsequent liberalizations and reform context

Read:

- Table AI shows that most countries did not complete liberalization in one step.
- Later openings include additional country funds, investability-index increases, higher foreign-ownership ceilings, and broader foreign access.
- The table also shows that later openings continue to be surrounded by macroeconomic and political events.

Economic message: Liberalization is a process rather than a single switch. This motivates distinguishing the first liberalization from all later liberalizations.

Table AI
Subsequent Stock Market Liberalizations and Contemporaneous Economic Reforms
T^{*} is the date of the country's stock market liberalization in event time. All events are taken from *The Economist Intelligence Unit: Quarterly Economic Reports*. A full chronology of events is presented in Henry (1999b).

Country, Opening Date	Type of Opening	<i>T</i> [*] - 12	<i>T</i> [*] - 9	<i>T</i> [*] - 6	<i>T</i> [*] - 3	<i>T</i> [*]	<i>T</i> [*] + 3
Argentina							
January 91	Investable Index jumps 19 percent	None	Airline and ship privatizations begin	Structural adjustment funds unimpaired	IMF agreement, privatizations	Domínguez Cavallo appointed finance minister	Tariff reductions
January 92	Country Fund	None	Privatizations	IMF stand by loan	None	IMF approves economic plan	IMF agreement; Brady deal
Brazil							
October 88	Country Fund	None	None	None	IMF approves economic program; import ban lifted	Creditors ratify new loan agreement	Third Crusado Plan
April 90	Investability Index jumps 33 percent	IMF talks open; stock market scandal	Tariffs reduced	Privatization process frozen	None	Collor takes office, sweeping deregulation	Tariffs reduced; curb on profit remittance removed
January 91	Investability Index jumps 34 percent	None	None	IMF talks open	Deregulation measures announced; debt restructuring rejected	Sarney Collor Plan	None
July 91	Investability Index jumps 185 percent;	None	None	None	Agreement on payment of arrears	IMF negotiations begin; privatizations	None
May 92	Country Fund	None	None	None	IMF approves a new stand by loan	Negotiations begin on Brady deal	Brady debt deal signed; official charges of corruption against Collor

continued

Table AI, part 1: Subsequent stock market liberalizations and reform events.

May 94	Country Fund	Government faces no confidence vote	None	None	None	Foreigners can enter telecom industry	None
September 94	Country Fund	None	None	None	None	None	None
Korea							
December 88	Government announces plan to open stock market	Rob Tae Woo elected president	Tariffs reduced on consumer durables	None	Minimum wage increased by 23 percent	Interest rates deregulated	Investment in foreign real estate allowed
July 90	Country Fund	None	None	None	North Korea proposes disarmament	Diplomatic relations with USSR	None
March 91	Country Fund	None	None	None	None	None	None
January 92	Foreigners allowed to hold up to 10 percent of market	None	None	None	Limit on foreign banks issue of cds eased	Bank bailout of \$680 million	North Korea agrees to military inspection
October 92	Investability Index jumps 23 percent	None	None	None	Pension funds urged to buy more equity	Kim Young Sam elected president	None
July 93	Country Fund	None	None	Governor of Bank of Korea is sacked	Financial reform plan published	Foreigners can buy convertible bonds	Real name financial system decree
December 93	Country Fund	None	None	None	Lending rates liberalized	GATT tariff reduction agreements	Foreign banks admitted
December 94	Foreign equity ceiling raised to 12 percent	None	Manufacturing firms can issue unlimited corporate funds	Kim Il Sung dies	None	None	None
Malaysia							
December 87	Country Fund	None	None	Possible cut in corporate tax rate announced	90 arrests under Internal Security Act	None	\$1 billion rescue plan for depositors
April 89	Country Fund	Most favored nation trade pact with China	None	ASEAN-Japan Development Fund loans	None	None	Huawei on restructuring foreign equity
April 90	Country Fund	None	Banks allowed to purchase stock	152 firms delist from Singapore	None	Plan for electric-ity privatization	None

Table AI, part 3: Continued subsequent liberalizations.

Table AI—Continued

Country, Opening Date	Type of Opening	<i>T</i> [*] - 12	<i>T</i> [*] - 9	<i>T</i> [*] - 6	<i>T</i> [*] - 3	<i>T</i> [*]	<i>T</i> [*] + 3
Chile							
June 88	Country Fund	None	None	Teléfonos de Chile privatized	Privatization of state electricity company begins	Poll shows Pinochet to win plebiscite	None
January 89	Investability Index jumps 15 percent	None	None	None	Pinochet defeated in plebiscite	None	None
February 90	Country Fund	None	IMF mission visits	IMF loan; Central Bank independent	Patricio Aylwin takes over as President	Foreign exchange controls eased	Aylwin announces commitment to reforms
January 91	Investability Index jumps 42 percent	None	None	Debt rescheduling	None	None	Capital outflow restrictions eased
January 92	Investability Index jumps 46 percent	None	None	Free trade agreement with Mexico	None	Peso revalued by 5 percent	Foreign exchange controls eased
India							
May 87	Country Fund	None	None	Stock market scandal	None	None	None
August 88	Country Fund	None	Talks on trade liberalization begin	Import liberalization package	Government declares support for privatization	None	None
December 88	Country Fund	None	None	None	None	None	None
October 89	Country Fund	None	None	None	None	Gandhi congress ousted	None
June 90	Country Fund	None	None	None	None	None	Import liberalization
May 92	Country Fund	Rao elected PM; rupee devalued	None	None	Exchange controls eased, import duties	Illegal stock trading exposed	None

Table AI, part 2: Continued subsequent liberalizations.

Country, Opening Date	Type of Opening	<i>T</i> [*] - 12	<i>T</i> [*] - 9	<i>T</i> [*] - 6	<i>T</i> [*] - 3	<i>T</i> [*]	<i>T</i> [*] + 3
Malaysia (continued)							
January 91	Investability Index jumps 29 percent	None	None	None	Prime Minister Mahdi's party retains power in general elections	None	None
Mexico							
October 90	Country Fund	Brady term sheet submitted	None	Privatization of banks approved	Salinas requests NAFTA talks; Trujillo to be privatized	None	None
January 92	Investability Index jumps 51 percent	None	NAFTA talks begin; EZB of Telmex privatized	Election: strong PRI showing boosts reforms	Bancomer privatized	None	Environmental concerns about NAFTA
The Philippines							
May 87	Country Fund	None	Import controls lifted	Paria Club debt rescheduling of \$870 million	\$10.5 billion structural adjustment loan; debt rescheduling	Agrarian land reform plan is approved	Coup attempt; bombings of businesses in Makati
November 89	Country Fund	IMF approves stabilization plan	None	Debt rescheduled; using \$2.2 billion	Brady deal reached in principle	Coup attempt	None
October 93	Country Fund	None	Airline privatization announced	IMF negotiations begin	Privatization of copper and shipyards	Privatization of steel company approved	IMF agreement reached
Taiwan							
December 86	Country Fund	None	None	None	Import tariffs reduced	None	Restrictions imposed on capital inflows
May 89	Country Fund	None	Capital gains tax imposed	Privatization of China Steel announced	More flexible exchange rate regime	Central bank governor resigns; trade restrictions	Exchange controls lifted; privatizations

Table AI, part 4: Continued subsequent liberalizations.

Table AII
Unique Stock Market Liberalization Dates

This table lists the unique liberalization dates from Table II. Column (1) lists all of the unique liberalization dates in Table II. Column (2) lists the unique liberalization dates from columns (3) through (5) of Table II.

								(1)	(2)
								All Unique Stock Market Liberalization Dates from Table II	Unique Stock Market Liberalization Dates from Table II, Columns (3)–(5) only
								Country	
								Argentina	November 1989 October 1991
								Brazil	March 1988 May 1991
								Chile	May 1987 September 1987 October 1989 January 1992
								Colombia	February 1991 October 1991 December 1991
								India	June 1986 November 1992
								Korea	June 1987 January 1992
								Malaysia	May 1987 December 1988
								Mexico	May 1989 November 1989
								The Philippines	May 1986 July 1986 June 1991 October 1989
								Taiwan	May 1986 January 1991
								Thailand	January 1988 September 1987
								Venezuela	January 1990 December 1988

Table AI, part 5: Continued subsequent liberalizations.

Table AII: Unique stock market liberalization dates.

6.7 Appendix Tables AIII and AIV: Alternative dates and later liberalizations

Read Table AIII:

- Table AIII repeats the analysis using alternative liberalization dates from the literature.
- The implementation-month-only return effect remains positive and statistically significant.
- After all controls, the coefficient is around 0.052 using the broad set of unique dates.

Read Table AIV:

- Table AIV separates the first liberalization from subsequent liberalizations.
- The first liberalization has a much larger return coefficient.
- *Liberalize2*, the dummy for later liberalizations, is small and statistically insignificant after controls.

Economic message: The result is not driven by one narrow date list, and the market reaction is concentrated in the first liberalization. Later reforms may be anticipated or may add little new diversification benefit.

Table AIII

Stock Market Reactions to First Stock Market Liberalization, Alternative Event Dates

The regressions are performed using monthly stock market data from December 1976 to December 1994 for Argentina, Brazil, Chile, India, Korea, Mexico, and Thailand. For the other countries the data are monthly from December 1984 to December 1994. The dividend yield data are also monthly and cover the period from December 1984 to December 1994. *Liberalize* is a dummy that takes on the value one during the implementation month of the first stock market liberalization. R^{IPC} , $R^{S\&P}$, and R^{MSVE} are the dividend-inclusive monthly returns on the IPC Global Index, the S&P 500, and the Morgan Stanley Capital Index for Europe, Asia, and the Far East, respectively. *Stabilize*, *Trade*, *Privatize*, and *Exchange* are dummy variables for the event windows of macroeconomic stabilization, trade opening, privatization, and exchange controls respectively. Each of the event windows for these variables begins seven months prior to the implementation of the reform and ends in the implementation month. A constant plus 11 country dummies were also estimated but not reported. Heteroskedasticity-consistent (White) standard errors are in parentheses.

	Panel A: Stock Returns					Panel B: $\Delta \ln(D/P)$				
	(1a)	(2a)	(3a)	(4a)	(5a)	(1b)	(2b)	(3b)	(4b)	(5b)
<i>Liberalize</i>	0.072*** (0.024)	0.057** (0.025)	0.056** (0.024)	0.052** (0.024)	0.051* (0.027)	-0.051* (0.029)	-0.041 (0.030)	-0.041 (0.030)	-0.034 (0.032)	-0.040 (0.041)
R^{LDC}		0.512*** (0.063)	0.507*** (0.062)	0.516*** (0.059)	0.519*** (0.143)		-0.343*** (0.103)	-0.334*** (0.103)	-0.334*** (0.104)	-0.335*** (0.116)
R^{US}		0.266*** (0.100)	0.272*** (0.100)	0.293*** (0.094)	0.293*** (0.108)		-0.363*** (0.140)	-0.372*** (0.140)	-0.452*** (0.156)	-0.453*** (0.205)
R^{EAFE}		-0.004 (0.036)	-0.002 (0.036)	-0.014 (0.033)	-0.015 (0.042)		-0.045 (0.055)	-0.047 (0.055)	-0.029 (0.056)	-0.029 (0.024)
<i>Stabilize</i>					0.004 (0.013)		0.003 (0.024)	0.003 (0.023)	0.003 (0.023)	0.003 (0.008)
<i>Trade</i>					0.025*** (0.008)		0.021*** (0.008)	0.021*** (0.005)	-0.039** (0.016)	-0.037** (0.017)
<i>Privatize</i>					0.017* (0.010)		0.011 (0.008)	0.011 (0.001)	-0.039* (0.018)	-0.039* (0.017)
<i>Exchange</i>					-0.007 (0.015)		-0.003 (0.014)	-0.003 (0.016)	0.008 (0.037)	0.008 (0.037)
R^2	0.001	0.006	0.070	0.147	0.145	0.000	0.010	0.010	0.030	0.030
No. of obs.	2292	2292	2292	2292	2292	1569	1569	1569	1569	1569

Table AIII: Alternative event dates.

Table AIV

Stock Market Reaction to First and All Subsequent Stock Market Liberalizations

The regressions are performed using monthly data from December 1984 to December 1994. *Liberalize* is a dummy variable that takes on the value one during the month that the first stock market liberalization is implemented. *Liberalize2* is a dummy variable that takes on the value 1 during the implementation month of all stock market liberalizations subsequent to the first. R^{IPC} , R^{US} , and R^{EAFE} are the monthly return on the IPC Global Index, the S&P 500, and the Morgan Stanley Capital Index for Europe, Asia, and the Far East, respectively. *Stabilize*, *Trade*, *Privatize*, and *Exchange* are dummy variables for the event windows of macroeconomic stabilization, trade opening, privatization, and exchange controls respectively. Each of the event windows for these variables begins seven months prior to the implementation of the reform and ends in the implementation month. A constant plus 11 country dummies were also estimated but not reported. Heteroskedasticity-consistent (White) standard errors are in parentheses.

	Panel A: Stock Returns				Panel B: $\Delta \ln(D/P)$			
	(1a)	(2a)	(3a)	(4a)	(1b)	(2b)	(3b)	(4b)
<i>Liberalize</i>	0.101*** (0.030)	0.082** (0.041)	0.078 (0.039)	0.066 (0.036)	-0.060 (0.049)	-0.043 (0.050)	-0.037 (0.049)	-0.003 (0.081)
<i>Liberalize2</i>		0.030 (0.022)	0.028 (0.021)	0.022 (0.018)		-0.056 (0.059)	-0.057 (0.060)	-0.074 (0.062)
R^{LDC}		0.520*** (0.150)	0.514*** (0.147)	0.524*** (0.143)		-0.353*** (0.120)	-0.343*** (0.116)	-0.325*** (0.115)
R^{US}		0.251*** (0.102)	0.258*** (0.101)	0.280*** (0.110)		-0.349* (0.185)	-0.359* (0.200)	-0.385** (0.191)
R^{EAFE}		-0.002 (0.044)	-0.001 (0.044)	-0.013 (0.042)		-0.049 (0.021)	-0.051 (0.022)	-0.041 (0.025)
<i>Stabilize</i>				0.005 (0.011)		0.003 (0.010)	-0.003 (0.011)	-0.001 (0.005)
<i>Trade</i>				0.025*** (0.005)		0.021*** (0.005)	-0.040*** (0.015)	-0.039** (0.017)
<i>Privatize</i>				0.016** (0.006)		0.010** (0.007)	-0.027 (0.019)	-0.026 (0.021)
<i>Exchange</i>				-0.007 (0.015)		-0.003 (0.016)	0.008 (0.037)	0.009 (0.046)
R^2	0.000	0.070	0.070	0.147	0.000	0.010	0.011	0.031
No. of obs.	2292	2292	2292	2292	1569	1569	1569	1569

Table AIV: First and subsequent stock market liberalizations.

7 Short summary

- The paper studies whether opening emerging stock markets to foreign investors causes an equity-price revaluation.
- The theory is that liberalization improves international risk sharing, lowers the cost of equity capital, and raises prices holding cash flows fixed.
- The sample covers 12 emerging markets in Latin America and Asia.
- The paper carefully dates the first liberalization using policy decrees, country funds, and instability-index jumps.
- Announcement dates from news databases are often unreliable, so the benchmark event window is $T^* - 7$ through T^* .
- The raw eight-month abnormal return is about 4.7 percent per month.
- After world-return controls, reform controls, and macroeconomic fundamentals, the estimate is about 3.3 percent per month, or about 26 percent cumulatively.
- Shorter windows are also positive; the implementation-month-only estimate is about 6.5 percent.
- Trade liberalization also has a large positive stock-price effect.
- Subsequent liberalizations have much smaller and statistically insignificant effects.
- The interpretation is consistent with a fall in the cost of equity capital, but causal attribution is complicated by simultaneous reforms and policy timing.

8 Conclusion

8.1 Core conclusion

Henry finds that emerging-market equity prices rise substantially in anticipation of initial stock market liberalizations. The evidence is consistent with international asset pricing models in which opening the market to foreign investors lowers the required return on equity.

8.2 Economic intuition

Before liberalization, domestic investors bear more of the local aggregate risk. After liberalization, foreign investors can share that risk. If the global price of risk is lower than the local price of risk, the equity premium falls. Lower discount rates raise asset prices even if expected cash flows do not change.

8.3 Incremental contribution revisited

The paper's main contribution is to connect stock market liberalization directly to equity-price revaluation while explicitly controlling for other policy reforms. It does not merely show that returns are high around liberalization; it asks whether the effect survives controls for global markets, trade reform, privatization, stabilization, exchange-control easing, and macro fundamentals.

8.4 Implications

The results suggest that financial globalization can have immediate valuation effects. If the cost of capital falls, more investment projects may become positive NPV, which provides a channel from liberalization to later investment growth and real economic activity.

8.5 Limitations

The event study cannot perfectly separate stock market liberalization from broader reform expectations. Liberalization may be timed to coincide with favorable economic news, and the policy itself may signal future reforms. The paper reduces these concerns but does not eliminate them completely.

8.6 Takeaway

The enduring takeaway is that the first opening of an emerging stock market to foreign investors is associated with a large equity-price revaluation. The result is strongest for the initial liberalization, remains positive under many controls and shorter windows, and is economically consistent with a lower cost of equity capital.